

Numerical Verification of Non-Abelian Braiding Statistics in Oxide Perovskite Topological Defect States

Input Question

Can topological quantum computing be realized?

Domain

Information Science

Validation Status

needs_data

Problem Statement

Whether the exploitation of topological phases and intermediate particles can effectively mitigate decoherence and environmental perturbations to realize a functional topological quantum computer.

Rationale

Traditional quantum computers suffer from decoherence due to external perturbations. Topological quantum computing (TQC) proposes using the geometric phases of particle groups (topologies) to encode information, potentially offering inherent protection against local noise. Evidence confirms the existence of various topological states of matter and identifies oxide perovskites as candidate materials for such applications. However, it remains unverified whether these specific material systems can support the non-Abelian braiding statistics required for universal TQC, or if they are limited to Abelian phases insufficient for fault-tolerant computation.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Oxide perovskite and double perovskite systems hosting realized topological defect states (e.g., Dirac vortices) are theoretically capable of supporting non-Abelian braiding statistics suitable for TQC, contingent upon specific spin-orbit coupling and symmetry-breaking conditions verifiable via material-specific numerical simulation.
- **Mechanism**: Evidence EV-Q094-e85864309cc5677951076f96 identifies oxide perovskites as candidates for quantum computing due to half-metallic ferromagnetism and topological properties. Evidence EV-Q094-61cbd9b7bb0439f2f0665c93 confirms the realization of defect states like Dirac vortices in topological materials. This hypothesis posits that when these specific material platforms host these specific defects, the resulting effective Hamiltonian can support non-Abelian zero modes, but this is a conditional theoretical prediction dependent on material parameters (e.g., SOC strength, magnetization direction) rather than an established fact. The mechanism links the material candidate (perovskite) to the computational resource (non-Abelian defect

mode) via a parameter-dependent topological phase transition.

- ****Falsifiable Prediction****: Numerical simulations using tight-binding or DFT-derived Hamiltonians specifically parameterized for oxide/double perovskites must demonstrate a topological phase transition to a non-Abelian phase (characterized by non-commutative braiding matrices or Chern number > 1) under realistic defect configurations. If simulations across the physically relevant parameter space for these perovskites only yield Abelian phases or trivial insulators despite the presence of Dirac vortices, the hypothesis that these specific materials are viable for non-Abelian TQC is falsified.
- ****Required Observations****: Construction of a defect-hosting Hamiltonian derived from oxide/double perovskite band structures (not generic Kitaev models) ; Numerical demonstration of non-commutative unitary transformations upon adiabatic exchange of defects in this specific material model ; Identification of the critical parameter window (e.g., SOC strength, exchange field) required to stabilize the non-Abelian phase in perovskites
- ****Risk of Being Wrong****: The 'topological properties' cited in EV-Q094-e85864309cc5677951076f96 may refer exclusively to Abelian topological phases (e.g., Quantum Anomalous Hall Effect) which, while useful for spintronics, do not support universal TQC. The realized Dirac vortices in EV-Q094-61cbd9b7bb0439f2f0665c93 may be inherently Abelian in the absence of specific superconducting proximity effects not guaranteed in intrinsic perovskites.

Hypothesis 2

- ****Hypothesis****: Current realizations of topological defect states in candidate materials represent Abelian topological phases insufficient for universal TQC without additional engineered proximity effects, implying that 'realization' of TQC requires hybrid heterostructures rather than intrinsic bulk perovskite properties.
- ****Mechanism****: EV-Q094-61cbd9b7bb0439f2f0665c93 lists 'fractionally charged modes' and 'helical modes' alongside Dirac vortices; these are frequently associated with Abelian statistics or time-reversal invariant topological insulators. EV-Q094-e85864309cc5677951076f96 highlights half-metallic ferromagnetism, which breaks time-reversal symmetry but does not guarantee p-wave pairing necessary for non-Abelian anyons. This hypothesis asserts that the intrinsic topological states verified in current evidence are computationally insufficient (Abelian), and TQC realizability depends on unverified interface engineering (e.g., inducing superconductivity) rather than the bulk material properties alone.
- ****Falsifiable Prediction****: If intrinsic perovskite defect states are sufficient for non-Abelian TQC, then isolated simulations of bulk perovskite defects (without external superconducting pairing terms) should exhibit non-Abelian degeneracy. Conversely, if simulations show only Abelian braiding statistics in isolated bulk models, and non-Abelian statistics emerge only upon adding explicit proximity-induced pairing terms, this supports the hypothesis that intrinsic properties are insufficient.
- ****Required Observations****: Comparative numerical analysis of braiding statistics in intrinsic vs. proximity-coupled perovskite defect models ; Verification that intrinsic defect modes correspond to Abelian representations of the braid group ; Demonstration that non-Abelian transitions require parameter regimes outside intrinsic material capabilities
- ****Risk of Being Wrong****: Intrinsic unconventional superconductivity or spin-triplet pairing might exist in certain double perovskites not captured by standard DFT functionals used in simulations, rendering the 'insufficiency' claim false. Alternatively, higher-order topology might enable computation through mechanisms distinct from standard non-Abelian braiding.

Technical Details

This experiment designs a numerical verification protocol to test the theoretical capability of oxide perovskites (EV-Q094-e85864309cc5677951076f96) hosting topological defect states (EV-Q094-61cbd9b7bb0439f2f0665c93) to support non-Abelian braiding statistics. Addressing the reviewer's 'Model-Material Mismatch', we replace generic Kitaev models with a material-specific tight-binding Hamiltonian derived from the electronic structure of half-metallic ferromagnetic double perovskites. The model incorporates strong spin-orbit coupling (SOC) and exchange fields characteristic of these oxides. Topological defects (Dirac vortices/disclinations) are introduced as spatial modulations in the hopping parameters or on-site potentials, consistent with the defect types confirmed in EV-Q094-61cbd9b7bb0439f2f0665c93. The core technical task is to simulate the adiabatic exchange of these defects and compute the resulting unitary transformation on the degenerate ground state manifold. We explicitly treat 'non-Abelian statistics' as a hypothesis to be verified via simulation, not an established fact, addressing the 'Evidence Gap'. The scope is restricted to numerical demonstration of non-commutative braiding matrices, aligning with the 'Scope Mismatch' critique by removing requirements for physical wet-lab demonstration.

Datasets

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"name": "Oxide Perovskite Electronic Structure Parameters",
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"description": "Tight-binding parameters (hopping integrals, SOC strength, exchange splitting) for half-metallic ferromagnetic double perovskites, derived from literature associated with EV-Q094-e85864309cc5677951076f96.",
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"evidence_ids": [
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"EV-Q094-e85864309cc5677951076f96"
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"access_status": "pending_download",
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"is_public_candidate": true
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"name": "Topological Defect Geometry Definitions",
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"description": "Geometric and potential profiles for Dirac vortices and disclinations in crystalline lattices, as realized in topological materials per EV-Q094-61cbd9b7bb0439f2f0665c93.",
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"evidence_ids": [
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"access_status": "pending_download",
"is_public_candidate": true
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## Target

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{
"name": "Perovskite Defect Braiding Simulation Results",
"description": "Numerical data containing ground state degeneracies, energy spectra during defect motion, and
computed unitary braiding matrices for oxide perovskite models.",
"format": "HDF5/JSON",
"metrics": [
"Non-Commutativity Metric ($||U_1 U_2 - U_2 U_1||$)",
"Ground State Degeneracy Scaling Factor",
"Topological Gap Minimum During Braiding"
]
}
'''
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## Paper Abstract

Background: Topological quantum computing promises inherent protection against decoherence by encoding information in global topological properties. While various topological states of matter have been reported, and oxide perovskites identified as promising candidates, the realization of a functional TQC device remains unproven. Methods: We propose a numerical study constructing material-specific tight-binding Hamiltonians for half-metallic ferromagnetic double perovskites, incorporating topological defects such as Dirac vortices. Validation Plan: We will simulate the adiabatic braiding of these defects and compute the resulting unitary operators to test for non-Abelian statistics. Results: pending (待执行验证实验). This study aims to bridge the gap between material candidates and computational viability.

## Methods

1. Material-Specific Hamiltonian Construction: Construct a multi-orbital tight-binding Hamiltonian for representative oxide double perovskites including SOC and exchange fields. 2. Defect Embedding: Introduce

topological defects by modifying lattice connectivity or on-site phases. 3. Exact Diagonalization/DMRG: Compute low-energy spectrum and identify degenerate ground state manifold. 4. Adiabatic Braiding Simulation: Simulate time-dependent evolution during defect exchange. 5. Unitary Matrix Extraction: Compute overlap matrix to extract braiding unitary operator. 6. Non-Abelian Verification: Calculate commutator of braiding operators to detect non-commutativity.

## Experiments

### Baselines

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"Trivial Insulator Model: A perovskite Hamiltonian without topological band inversion or SOC, expected to show no protected zero modes or degenerate ground states.",

"Abelian Topological Phase Model: A quantum anomalous Hall (QAH) model parameterized for perovskites, expected to host chiral edge states but Abelian anyonic excitations (commutative braiding phases).",

"Generic Kitaev Honeycomb Model: A standard toy model for non-Abelian anyons, used only as a theoretical reference for ideal non-Abelian behavior, distinct from the material-specific test."

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### Metrics

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"Non-Commutativity Norm: $||[U_{ij}, U_{jk}]||$, where U_{ij} is the unitary for exchanging defects i and j . Zero for Abelian/trivial, non-zero for non-Abelian.",

"Ground State Degeneracy: Number of zero-energy states as a function of defect count N . Expected scaling $\sim 2^{N/2}$ for Ising-like non-Abelian anyons.",

"Topological Gap Robustness: Minimum energy gap between the ground state manifold and the first excited state during the entire braiding process.",

"Fidelity to Theoretical Braid Group: Overlap between simulated unitary matrices and the theoretical representation of the braid group for non-Abelian anyons."

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### Ablation

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"SOC Strength Variation: Vary spin-orbit coupling strength to determine the threshold required for the topological phase transition in the perovskite model.",

"Defect Separation Distance: Test the stability of zero-mode localization and braiding fidelity as a function of distance between defects.",

"Disorder Introduction: Add random on-site potential disorder to test the robustness of the topological protection against material imperfections common in oxides."

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Validation Protocol

1. Verify that the bulk Hamiltonian exhibits a non-trivial topological invariant (e.g., Chern number or Z2 index) in the clean limit.
2. Confirm that introducing defects creates localized zero-energy modes within the bulk gap.
3. Demonstrate that the ground state degeneracy increases with the number of defects.
4. Show that the braiding unitary matrices do not commute (non-Abelian signature) and differ significantly from the Abelian baseline.
5. Ensure the topological gap remains open throughout the braiding path to validate adiabaticity.

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- **EV-Q094-61cbd9b7bb0439f2f0665c93** · arxiv · arXiv:2208.05082
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2208.05082.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:2|section:page-2|paragraph:1;
content_sha256=88dd73a84f4e9695efd81bb893055a416ce88f7e39c6d8ccae7dba16c03eac2c
- **EV-Q094-4067c58c349416aacf234ada** · arxiv · arXiv:2409.00963
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2409.00963.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:3|section:page-3|paragraph:1;

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• **EV-Q094-e85864309cc5677951076f96** · arxiv · arXiv:2208.11988

• authors: Not available · year: Not available

• url: <https://arxiv.org/pdf/2208.11988.pdf> · doi: Not available

• reliability_note: eligibility_status=FULLTEXT_VERIFIED;

topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;

content_sha256=cd47b7a09f60a1c79bce16fe6d1d6f85d949e5cd1f332743efcdffd5212cc24f

Reviewer Comments

- The revised hypothesis correctly reframes non-Abelian statistics as a conditional theoretical prediction contingent on material-specific parameters, resolving the previous evidence gap regarding established facts.
- Experiment design now explicitly targets oxide perovskite Hamiltonians derived from EV-Q094-e85864309cc5677951076f96, successfully closing the model-material mismatch identified in iteration 1.
- Falsifiable predictions and required observations are now strictly aligned with the numerical simulation scope, removing unsupported claims of physical wet-lab demonstration.
- Knowledge gaps in EvidenceExtractionResult have been appropriately updated to reflect that non-Abelian nature is unverified in current evidence cards.
- Results field correctly maintains 'pending' status without fabrication.

Revision History

- auto_revision_1: 依据评审意见重做假设与实验设计。

Reproducibility Checklist

- Source code for constructing the oxide perovskite tight-binding Hamiltonian is version-controlled and documented.
- Parameters for SOC, exchange field, and hopping integrals are explicitly listed and traced to EV-Q094-e85864309cc5677951076f96.
- Defect insertion algorithm follows the geometric descriptions in EV-Q094-61cbd9b7bb0439f2f0665c93.
- Random seeds for any disorder realization are fixed and recorded.
- Convergence criteria for DMRG/ED calculations (bond dimension, truncation error) are specified.
- Scripts for computing braiding unitaries and non-commutativity metrics are provided.